Physics

Answer Key

1. MECHANICS

1. The time of the meeting was 8 a.m. The man's speed was 4 km/h. The other questions can easily be answered with the aid of the chart in Fig. 1 – the man met the twelfth bus at a distance of 10.7 km from the mill; the cyclist was overtaken by four buses.

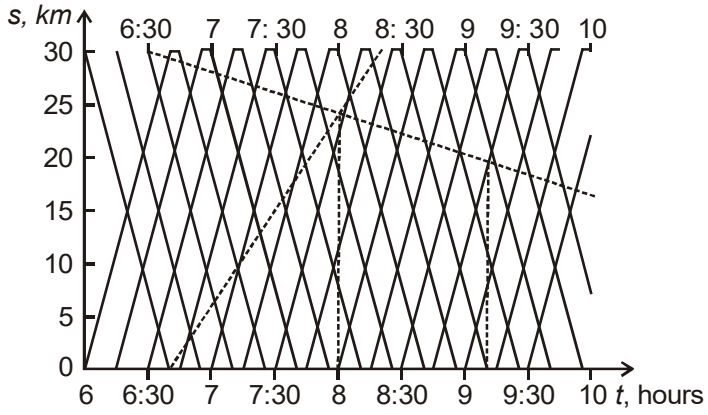


Fig. 1

2. Each tourist will walk half the distance and ride the bicycle the other half. The entire distance will be covered in $t = \frac{s}{2v_1} + \frac{s}{2v_2}$ = 5 hours 20 minutes.

   Hence, the mean speed is $v = \frac{s}{t} = 7.5\,\text{km/h}$. The bicycle remains unused during half the time of motion, *i.e.*, during 2 hours 40 minutes.

3. It is therefore to cover the distance *AB* in 30 minutes.

4. $v = u\,\frac{\sin\left(\alpha + \beta - \frac{\pi}{2}\right)}{\sin(\pi - \beta)}$.

   It is impossible to determine the velocity of the current from the known speed of the launch with respect to the bank, since we do not know the direction of the moving launch with respect to the water.

5. $v = \frac{1}{\sin\alpha}\sqrt{v_1^2 + v_2^2 + 2v_1v_2\cos\alpha}$

6. The charts show different motions.

7. The time during which the first ball falls is $t_1 = \sqrt{\frac{2h_1}{g}} = 0.3$ s. The ratio of the maximum velocities of the balls is $\frac{v_2}{v_1} = \sqrt{\frac{h_2}{h_1}} = \frac{1}{2}$.

It follows from the velocity chart (Fig. 7) that the mmimum time τ = 0.3 s. Besides. the second ball may begin to fall in 0.6, 0.9. 1.2 s, etc., after the first ball begins to drop.

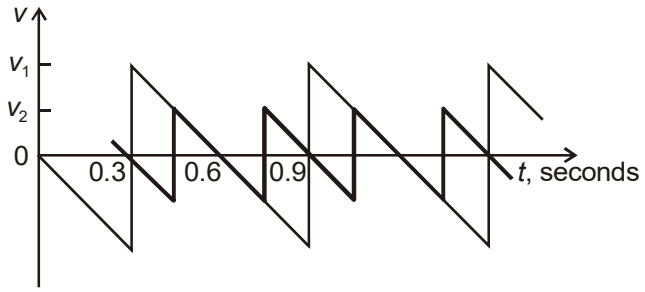


Fig. 7

The time t during which the velocities of the two balls are the same is 0.3 s. The process periodically repeats every 0.6 second.

8. The time during which the load is lifted to the height h is $t = \sqrt{\frac{2h}{a_1}}$. The speed of the load relative to the crane in a vertical direction at this moment is $v_1 = a_1 t$ and in a horizontal one $v_2 = a_2 t$.

The total speed of the load with respect to the ground is

$$v = \sqrt{v_1^2 + v_2^2 + v_0^2}$$

9. The dynamometer first shows F = 3 kgf.

G_1 = 3 kgf.

10. In 5 seconds the cart will stay at the same place (x = 0) and will have a speed v_t = 7 m/s directed to the right. The cart will cover the distance

$$s = 2\left\{v_0 \frac{t}{2} - \frac{a\left(\frac{t}{2}\right)^2}{2}\right\} = 17.5 \text{ meters}$$

11. $$a = \frac{(m_1 - m_2)\, g - F}{m_1 + m_2}$$

12. (1) $a_1 = a_2 = \frac{F}{G_1 + G_2}\, g = \frac{8}{25} g$

(2) $a_1 = \frac{-F + kG_2}{G_1}\, g = -\frac{2}{15} g$ and it will move to the left.

13. The speed of the cart will not depend on the point of impact. The momentum of the revolving cylinder is zero, irrespective of its direction and velocity of rotation. For this reason the bullet will impart to the "cylinder-cart" system the same momentum as it would to a cylinder rigidly secured on the cart.

14. For $\frac{m_1}{m_2} \le \sin\alpha - k\cos\alpha$, the wedge will move to the left

For $\frac{m_1}{m_2} \ge \sin\alpha + k\cos\alpha$, the wedge will move to the right.

For the wedge to be in equilibrium, the ratio between the masses of the weights should satisfy the inequality $\sin\alpha - k\cos\alpha \le \frac{m_1}{m_2} \le \sin\alpha + k\cos\alpha$

15. The sought distance $h = \ell \cot\alpha = \frac{\ell}{\sqrt{3}}$. Equilibrium will set in after the oscillations caused by the weight being lowered attenuate.

16. $\tan\alpha \le k$.

17. If at the moment when end *B* of the rod begins to rise the force of friction $F_{fr} \le kN$ proves sufficient for end *A* not to slip, the rod will begin to rotate around point *A*. Otherwise, end *A* will begin to slip until the force of friction $F_{fr} = kN$ can keep the rod in equilibrium

18. The ladder cannot be prevented from falling down by means of a rope tied to its middle.

19. $k \ge \frac{\ell\cos\alpha + 2\sqrt{d^2 - l^2\sin^2\alpha}}{\ell\sin\alpha}$

20. $F_{\min} = \frac{kmg}{\sqrt{1+k^2}}$.
$\tan\alpha_1 = k$.

21. $CO = \frac{4}{3}\frac{\sin\frac{\alpha}{2}}{\alpha}r$ from point *O*.

22. $u_1 = u_2 = \frac{v}{2}$.

23. The striking ball will jump back and the other balls up to the steel one will remain in place. The steel ball and all the others after it will begin to move to the left with different velocities.

The fastest velocity will be imparted to the extreme left-hand ball. The next one will move slower, etc. The balls will move apart.

24. $v = \sqrt{g\ell}$

25. $v^2 = \frac{2m^2g\ell}{(M+m)} \cdot \frac{(\cos\beta - \cos\alpha)\cos^2\beta}{(M + m\sin^2\beta)}$

26. $a = \frac{gh\sqrt{h^2 + 4\pi^2R^2 + 64\pi^4n^2R^2}}{h^2 + 4\pi^2R^2}$

27. The instantaneous axis of rotation passes through point *C* (Fig. 19). For this reason the velocity of point *A* relative to the block is $v_A = v\frac{R+r}{r}$.

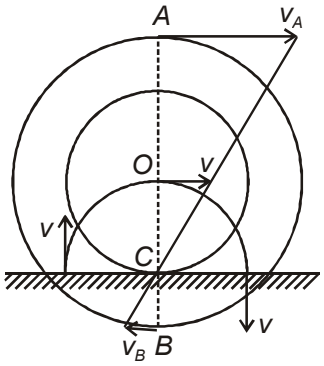


Fig. 19

Point *B* has the velocity $v_B = v\dfrac{R+r}{r}$. Point on a circle with the radius *r* whose centre is point *C* have an instantaneous velocity equal to that of the spool core.

28. $v_0 = \dfrac{1}{2}(v_1 + v_2) = \dfrac{1}{4}(\omega d + \Omega D)$

In this expression, each angular velocity may be positive (clockwise rotation) and negative (counterclockwise rotation). When $\Omega = 0$, $v_0 = \dfrac{\omega d}{4}$.

29. $\omega = \dfrac{v_1 - v_2}{\ell}$, and $u = \dfrac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$

30. $\sqrt{gR\left(2 + 2\cos\alpha + \dfrac{1}{\cos\alpha}\right)}$

31. $\alpha = 60°$.

It is easy to see that if $\alpha > 60°$ the body will fall inside the loop; if $\alpha < 60°$ it will fly out.

32. $v_1 = 2r_1 \sin\dfrac{\alpha}{2}\sqrt{g\dfrac{m_1 r_1 + m_2 r_2}{m_1 r_2^2 + m_2 r_2^2}}$

$v_2 = 2r_2 \sin\dfrac{\alpha}{2}\sqrt{g\dfrac{m_1 r_1 + m_2 r_2}{m_1 r_2^2 + m_2 r_2^2}}$

33. $m\sqrt{g^2 + \omega^4 \ell^2 \sin^2\phi}$

34. (1) $\phi = 0$, and (2) $\cos\phi = \dfrac{g}{\omega^2}\dfrac{mb + M(a+b)}{mb^2 + M(a+b)^2}$

The first solution is true for any angular velocities of rotation, and the second when $\omega \geq \sqrt{g\frac{mb + M(a+b)}{mb^2 + M(a+b)^2}}$.

35. In the state of equilibrium $m\omega^2 x = kx$, where x is the distance from the body to the axis.

It is thus obvious that with any value of x the spring imparts the centripetal acceleration necessary for rotation to the body. For this reason the latter will move after the impetus with a constant velocity up to stop *A* or as long as the law of proportionality between the force acting on the spring and its deformation is valid.

36. The angle α at which the motion of the spool is reversed can be found from triangle *AOB*; namely, $\sin\alpha = \frac{r}{R}$. If the thread is inclined more than α, the spool will roll to the right, If α is smaller it will move to the left on condition that it does not slip. If the tension of the thread *T* satisfies the condition $Tr \leq fR$, where *f* is the force of friction, the spool will remain in place. Otherwise, when $\sin\alpha = \frac{r}{R}$ it will begin to rotate counterclockwise around point *O*.

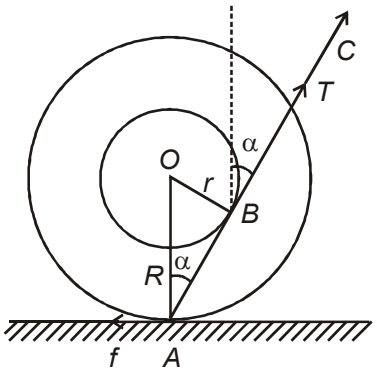


Fig. 25

37. Hence, the acceleration of the weight is

$$a = \frac{mg}{m + 2M\left(\frac{R}{R-r}\right)^2}$$

The sought tension *T* is

$$T = \frac{2m\,Mg\left(\frac{R}{R-r}\right)^2}{m + 2M\left(\frac{R}{R-r}\right)^2}$$

$$F = \frac{Mmg\left(1+\frac{r}{R}\right)}{m\left(1-\frac{r}{R}\right)^2 + 2M}$$

$$k < \frac{1+\frac{r}{R}}{2\frac{M}{m}+\left(1-\frac{r}{R}\right)^2}$$

38. $F = \frac{4}{3}\pi\gamma m\rho\sqrt{\frac{R^6}{\ell^4}+\frac{r^6}{\left(\ell^2-d^2\right)}-\frac{2R^3r^3\cos\beta}{\ell^2\left(\ell^2-d^2\right)}} \cong 5.8\times10^{-4}$ gf

39. The force of gravity imparts the same acceleration to the pendulum and the block. Gravity does not cause any deformations in the system during free falling. For this reason the pendulum will so move with respect to the block as if there is no gravitation. The pendulum will move with a constant angular velocity as long as the block falls.

40. The field of gravity is smaller near the bubble than in a homogeneous liquid, and the liquid is compressed less. For this reason one bubble will move into the volume of liquid near the other one, and the bubbles will be mutually attracted.

Two bubbles in a homogeneous liquid with negligibly small masses can be considered from a formal standpoint as negative masses superimposed upon the positive mass m of the medium in the volume of a bubble:

$$F = \gamma\frac{(-m)(-m)}{R^2} = \gamma\frac{m^2}{R^2}$$

41. $x = \frac{\gamma_0 h}{\gamma_0-\gamma} = 50$ cm

42. The resulting pressure at point A is $p = \rho(gh + a\ell)$.

2. HEAT MOLECULAR PHYSICS

1. $\alpha \cong 2.3\times10^{-5}$ deg^{-1}

2. $\theta = \frac{(m_1t_1+m_2t_2)c+qt_1}{(m_1+m_2)c+q} \cong 4°C$

3. $T_2 = \frac{\lambda_2d_1T_0+\lambda_1d_2T_1}{\lambda_2d_1+\lambda_1d_2}$

4. $\lambda = \frac{\lambda_1+\lambda_2}{2}$

5. $T_{max} \cong 490°K$

6. $p = -\rho gh$.

7. The amount of heat that can be liberated by the water when it is cooled to 0°C is 4,000 cal. Heating of the ice to 0°C requires 12,000 cal. Therefore, the ice can be heated only by the heat liberated when the water freezes. One hundred grammes of water should be frozen to produce the lacking 8,000 calories.

As a result, the calorimeter will contain a mixture of 500 g of water and 500 g of ice at a temperature of 0°C.

8. $\alpha = \sqrt[3]{\frac{Mg}{AE}}$

9. $F_c = \frac{mg}{4} = 25$ kgf and $F_s = 2F_c = 50$ kgf.

3. ELECTRICITY AND MAGNETISM

1. $C_0 = 1.2C$.
2. Since the dielectric is polarized, the intensity will increase at points *A* and *C* and decrease at point *B*.
3. $C = \frac{A}{4\pi d}\left\{1+(\varepsilon_r - 1)\frac{\ell_1}{\ell}\right\}$

   $C = \frac{\varepsilon_r A}{4\pi\{d_1 + \varepsilon_r (d - d_1)\}}$

4. $R_{AB} = \frac{(r_a + r_b)r_c}{r_a + r_b + 2r_c}$

   where $r_a = \rho\frac{a}{A}, r_b = \rho\frac{b}{A}, r_c = \rho\frac{\sqrt{a^2 + b^2}}{A}$

   $R_{CD} = \frac{2r_a r_b + r_c(r_a + r_b)}{r_a + r_b + 2r_c}$

5. $I = -\frac{\Delta q_1}{\Delta t} = \frac{\Delta q_2}{\Delta t} = \frac{Qv}{2d_0}$
6. The sought resistance is $R = 7\ \Omega$.
7. $P_0 = P\{1+\alpha(t - t_0)\} \cong 600$ watts
8. $v = \sqrt{\frac{2Q^2}{mR}}$
9. The field at the arbitrary point on *AB* induced by the entire conductor is zero
10. The intensity $H = k\cdot 2\pi jd$ is the same for all the points of the space.
11. $W = 2kHIab$.
12. $\phi_A - \phi_B = kHa\frac{I}{neA}$.
13. $\sigma = \frac{1}{4\pi}kvH$.
14. When the circuit moves, the magnetic flux passing through area *ABCD* diminishes. Therefore, in accordance with Lenz's law, the induced current will flow clockwise.
15. $I = 10^{-8}\frac{kh}{4\rho}(b^2 - a^2)$
16. $I_1 = \frac{10^{-8}ka}{r}\times\frac{7}{22}, I_2 = \frac{10^{-8}ka}{r}\times\frac{3}{11}$, and $I_3 = \frac{10^{-8}ka}{r}\times\frac{1}{22}$

17. $I_3 = \dfrac{\varepsilon R_1}{R_1R_2 + R_1R_3 + R_2R_3}$

18. $I_1 = 0, I_2 = \dfrac{\varepsilon}{R_2} = I_3$

$$U_A - U_B = -\frac{\varepsilon R_1}{R_1 + R_2}$$

Here $U_A - U_B = -\varepsilon_1$ (since $I_1 = 0$ on section *ADB*), where ε_1 is the e.m.f. of induction on section *ADB*.

19. It is easy to see that as the velocity increases, the acceleration *a* will diminish and become zero at the moment of equality of the forces $f = mg$. From this moment on, the conductor will move with a constant velocity v_k equal to

$$v_k = \frac{mgR \times 10^9}{H^2\ell^2}$$

20. $a = \dfrac{F}{m + 10^{-9}H^2\ell^2C}$

21. In both cases equilibrium will set in if the moment of the forces acting from the magnetic field on the current induced in the ring is equal to zero, or if there is no induced current. This will occur only if the plane of the ring is arranged along the force lines of the field (the induced current is zero) or when the plane of the ring is strictly perpendicular to the force lines (the moment of the forces is zero).

According to Lenz's law, the first position of the ring will be stable in an increasing magnetic field, and the second unstable.

In a decreasing magnetic field, on the contrary, equilibrium will be stable when the angle between the plane of the ring and the force lines is a right one, and unstable when the plane of the ring is parallel to the force lines.

22. $I = \dfrac{\pi r^2 H_0}{L}$.

4. OSCILLATIONS AND WAVES

1. $T = 2\pi\sqrt{2\dfrac{R-r}{g}}$

2. $T = 2\pi\sqrt{\dfrac{m_1\ell_1^2 + m_2\ell_2^2}{m_1\ell_1 + m_2\ell_2}\dfrac{1}{g}}$

3. $T = 2\pi\sqrt{\dfrac{\pi r}{2g}}$

4. $T = 2\pi\sqrt{\dfrac{m\ell}{4f}}$

5. Let us deflect both pendulums from the vertical in the same direction and through the same angle. In this case the spring will not be deformed. It is easy to see that the pendulums released from this position will oscillate in phase with a frequency of $\omega = \sqrt{\frac{g}{\ell}}$.

$$\omega = \sqrt{\frac{g + \frac{2k\ell}{m}}{\ell}}$$

5. GEOMETRICAL OPTICS

1. When the mirror turns through an angle α the reflected ray will turn through 2α since the angles of incidence and reflection increase by α. Hence, the angular velocity of rotation of the reflected ray is $\omega = 2\pi n \times 2$. The linear velocity with which the light spot moves along the screen is $v = 4\pi nR \cong 62.8$ m/s.

2. One-sixth of all the energy of the beam will get into the reveiver during one revolution, which is a sufficiently large interval of time.

3. The amount of water required is $V = xa^2 \cong 43.2$ litres.

4. $$f_1 = \frac{(n-1)f}{\frac{n}{n_1} - 1} = 90 \text{ cm}$$

$$f_2 = \frac{(n-1)f}{\frac{n}{n_1} - 1} = -102 \text{ cm}$$

5. $$n_2 = \frac{fDn_1}{fD + 1 - n_1} \cong 1.67$$

6. The centre of the lens O is the point of intersection of straight lines SS' and N_1N_2. The foci can easily be found by constructing the rays parallel to the major optical axis (Fig. 5).

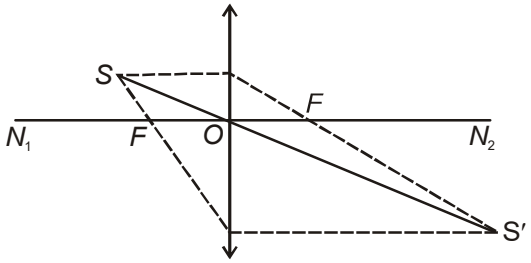


Fig. 5

7. $H = H' + d\left(1 - \frac{1}{n}\right) = 85$ cm.

8. The light beam issuing from a point at a distance of a_2 = 5 cm from the second lens impinges on it. As follows from the formula of a lens, the continuations of the rays refracted by this lens intersect at a distance of b_2 = 4 cm from it (Fig. 7). This point coincides with the focus of the third lens. For this reason the rays leaving

the system will travel in a parallel beam. The given system is telescopic.

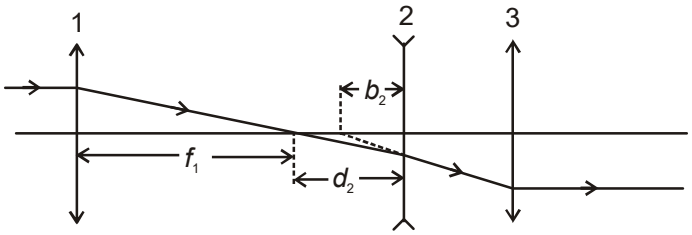


Fig. 7

9. $\frac{1}{a_3} = \frac{1}{a_1} + \frac{1}{f_0} = \frac{1}{a_1} - \frac{1}{a_2}$

6. PHYSICAL OPTICS

1. $\Delta h = \frac{\lambda(a+b)}{2b\alpha}$

2. The second coherent source is obtained in Lloyd's experiment by reflection of the rays from mirror *AO*. In reflection the phase changes by π (loss of a half-wave) and the oscillations will be damped at point *O* where a bright band should be observed (a minimum of illumination). The entire pattern will be shifted by the width of a light (or dark) band.

3. The interference bands will have the form of concentric circles.

$$h_{n-1} \cong \sqrt{\frac{2D(D+n\lambda)}{n}} = \sqrt{2D\lambda\left(\frac{D}{l}+1\right)}$$

4. $\Delta h = \lambda D = \frac{\lambda(a+b)}{2a\alpha(n-1)} = 0.15 \text{ cm}$

5. $\delta = \beta\frac{n_2 - n_1}{n_2 - 1}$

6. The lens is too thick. Interference occurs only with thin films. The air layer between the lens and the glass is thin.

7. $r_k = \sqrt{\frac{k\lambda}{\left(\frac{1}{R_2} - \frac{1}{R_1}\right)}}$

❑ ❑ ❑

Chemistry

Answer Key

1. Answer (D)
2. Answer (B)
3. Answer (C)
4. Answer (B)
5. Answer (C)
6. Answer (D)
7. Answer (C)
8. Answer (A)
9. Answer (B)
10. Answer (D)
11. Answer (C)
12. Answer (A, C)
13. Answer (A, B, C)
14. Answer (A, B, C)
15. Answer (B, C)
16. Answer (B)
17. Answer (B)
18. Answer (D)
19. Answer (B)
20. Answer (C)
21. Answer (C)
22. Answer (A, B, C)
23. Answer (A, B, C)
24. Answer (A, C)
25. Answer (A, C, D)
26. Answer A $\rightarrow$ (P); B $\rightarrow$ (S); C $\rightarrow$ (R); D $\rightarrow$ (Q, T)
27. Answer (16)
28. Answer (1)
29. Answer (B)
30. Answer (C)
31. Answer (B)
32. Answer (B)
33. Answer (D)
34. Answer (B)
35. Answer (C)
36. Answer (C)
37. Answer (B)
38. Answer (B)
39. Answer (D)
40. Answer (A)
41. Answer (D)
42. Answer (D)
43. Answer (D)
44. Answer (B)
45. Answer (D)
46. Answer (B, D)
47. Answer (9)
48. Answer (28)
49. Answer (16)
50. Answer (03)
51. Answer (D)
52. Answer (B)
53. Answer (B)
54. Answer (C)
55. Answer (A)
56. Answer (C)
57. Answer (B)
58. Answer (D)
59. Answer (A, B, D)
60. Answer (A, B, D)

61. Answer (A, B, D)
62. Answer (B, D)
63. Answer (B, C)
64. Answer (41)
65. Answer (1)
66. Answer (B)
67. Answer (B)
68. Answer (B)
69. Answer (C)
70. Answer (B)
71. Answer (B)
72. Answer (B)
73. Answer (C)
74. Answer (D)
75. Answer (B)
76. Answer (D)
77. Answer (A $\rightarrow$ P, Q, R); (B $\rightarrow$ P, Q); (C $\rightarrow$ S, T); (D $\rightarrow$ P, S)
78. Answer (B)
79. Answer (C)
80. Answer (B)
81. Answer (B)
82. Answer (B)
83. Answer (A)
84. Answer (D)
85. Answer (C)
86. Answer (A)
87. Answer (B)
88. Answer (C)
89. Answer (D)
90. Answer (D)
91. Answer (C)
92. Answer (A)
93. Answer (B)
94. Answer (C)
95. Answer (B)
96. Answer (A)
97. Answer (A)
98. Answer (C)
99. Answer (C)
100. Answer (D)
101. Answer (A)
102. Answer (A, B, C)
103. Answer (A, B, C)
104. Answer (A $\rightarrow$ Q); (B $\rightarrow$ P); (C $\rightarrow$ S); (D $\rightarrow$ R)
105. Answer (1)
106. Answer (9)
107. Answer (6)
108. Answer (C)
109. Answer (C)
110. Answer (B)
111. Answer (C)
112. Answer (D)
113. Answer (C)
114. Answer (A)
115. Answer (D)
116. Answer (C)
117. Answer (C)
118. Answer (D)
119. Answer (C)
120. Answer (B)
121. Answer (A, B, C, D)
122. Answer (C)
123. Answer (A, D)
124. Answer (0)
125. Answer (C)
126. Answer (A)
127. Answer (B)
128. Answer (A)
129. Answer (D)
130. Answer (D)
131. Answer (D)
132. Answer (A)
133. Answer (A)
134. Answer (C)
135. Answer (B)
136. Answer (C)
137. Answer (B, C)
138. Answer (3)

❑ ❑ ❑

MATHEMATICS

Answer Key

1.	(3)	2.	(2)	3.	(1)	4.	(2)	5.	(2)	6.	(2)
7.	(3)	8.	(3)	9.	(2)	10.	(1)	11.	(1)	12.	(1)
13.	(1)	14.	(?)	15.	(2, 4)	16.	(1, 2)	17.	(3, 4)	18.	(1)
19.	(1)	20.	(2)								
21.	(1) → (a); (2) → (a); (3) → (c) : (4) → (d)										
22.	(148)	23.	(96)	24.	(2)	25.	(1)	26.	(2)	27.	(3)
28.	(1)	29.	(3)	30.	(1)	31.	(2)	32.	(2)	33.	(1)
34.	(1)	35.	(2)	36.	(1)	37.	(1)	38.	(1)	39.	(3)
40.	(4)	41.	(1)	42.	(3)	43.	(2)	44.	(1, 2, 3, 4)	45.	(1, 2, 3, 4)
46.	(1, 2)	47.	(1, 2)	48.	(1, 2, 3, 4)	49.	(1, 4)	50.	(1, 2, 3, 4)	51.	(1, 2, 3)
52.	(1, 2, 3, 4)	53.	(3)	54.	(2)	55.	(3)	56.	(1)	57.	(2)
58.	(3)										
59.	(1) → (c); (2) → (a, b); (3) → (a, c); (4) → (d)										
60.	(3)	61.	(4019)	62.	(1)	63.	(4)	64.	(48)	65.	(??)
66.	(1)	67.	(4)	68.	(2)	69.	(3)	70.	(4)	71.	(1)
72.	(1)	73.	(2)	74.	(1)	75.	(3)	76.	(3)	77.	(1)
78.	(3)	79.	(3)	80.	(2)	81.	(4)	82.	(3)	83.	(1, 2, 3)
84.	(1, 2)	85.	(2, 4)	86.	(1, 3, 4)	87.	(2)	88.	(3)	89.	(3)
90.	(3)	91.	(4)	92.	(2)						
93.	(1) → (b); (2) → (c); (3) → (a), (c); (4) → (d)										
94.	(2)	95.	(1)	96.	(1)	97.	(104)	98.	(3)	99.	(3)
100.	(2)	101.	(2)	102.	(2)	103.	(1)	104.	(4)	105.	(3)
106.	(4)	107.	(2)	108.	(3)	109.	(4)	110.	(3)	111.	(1)
112.	(1, 3)	113.	(2, 3, 4)	114.	(1, 3)	115.	(4)	116.	(4)	117.	(2)
118.	(?)	119.	(4)								
120.	(1) → (a, d), (2) → (b), (3) → (b, d), (4) → (c)										
121.	(5)	122.	(6)	123.	(16)	124.	(2)				

❑ ❑ ❑